

Project1_Final_Submission

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1 STAT 5243 Project 1: WallStreetBets Data Pipeline and Exploratory Analytics

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Reviewer Contract: This notebook is written so a professor can grade from **one PDF and one code file**. - Single report PDF target: Project Deliverables/Report/STAT5243_Project1_Team2_Final.pdf - Single canonical workflow code file: Project Deliverables/Code Files/Project1_Full_Workflow_Code.ipynb - Backup automation script: Project Deliverables/Code Files/full_workflow.py

1.1 1. Introduction and Dataset Description

This project analyzes **53,187 posts** from `r/wallstreetbets` during the 2020-2021 meme-stock period. The dataset is complex because it combines: - structured engagement variables (`score`, `comms_num`, timestamps, URLs), - unstructured language data (`title`, `body`) with mark-down/slang/noise, - temporal behavior with regime-shift dynamics.

Quantitative complexity indicators include: - `missing_body_pct` = 53.49% (structural missingness), - heavy-tailed engagement distributions with viral outliers, - mixed raw/derived feature requirements for modeling readiness.

These characteristics justify advanced-level cleaning, EDA, and feature-engineering work.

```
[ ]: import pandas as pd

raw = pd.read_csv('../Datasets/reddit_wsb.csv')
clean = pd.read_csv('../Datasets/reddit_wsb_cleaned.csv')

print('raw_shape:', raw.shape)
print('clean_shape:', clean.shape)
print('raw_columns:', list(raw.columns)[:8], '...')
```

1.2 2. Data Acquisition Methodology

The dataset was acquired from a **public repository (Kaggle)** and originally collected from Reddit via **PRAW**. We preserve the raw file unchanged and perform deterministic transformation into the cleaned analytical dataset.

1.2.1 Acquisition workflow

1. Download public dataset and preserve immutable raw input.
2. Validate schema and coverage before transformation.
3. Apply deterministic cleaning/preprocessing workflows.
4. Export cleaned dataset and evidence artifacts (figures + JSON) for full auditability.

1.2.2 Why one-source can still be advanced

Although this project uses a single primary source, the source itself has significant real-world quality challenges (missing text, skewed targets, mixed structures), which creates advanced-level data engineering and interpretation work.

Evidence sources: - Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_policy_metrics.json - Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig_01_missingness_di

```
[ ]: # Integrity checks for acquisition -> processing continuity
print('row_count_raw:', len(raw))
print('row_count_clean:', len(clean))
print('same_row_count:', len(raw) == len(clean))
```

1.3 3. Cleaning and Preprocessing Steps

Cleaning is implemented as policy-driven transformation with explicit handling of each required inconsistency class.

1.3.1 3.1 Type consistency and formatting normalization

- Datetime coercion and temporal normalization.
- Text normalization for markdown/URL cleanup.
- Uniform transformed fields (score_log, comms_num_log, normalized variants).

1.3.2 3.2 Duplicate handling

- Duplicate checks on IDs and full rows.
- Result: no duplicate IDs and no duplicate full rows.

1.3.3 3.3 Missing data handling

- Structural missingness retained with explicit indicator variables instead of naive row deletion.
- Body missingness quantified and documented.

1.3.4 3.4 Outlier strategy

- Heavy-tailed variables are stabilized using log1p transforms.
- Diagnostics evaluate post-transform behavior and leverage influence.

1.3.5 3.5 Scaling and encoding

- Z-score and min-max scaling for numerical comparability.

- Categorical encoding and grouped post-type features for modeling inputs.

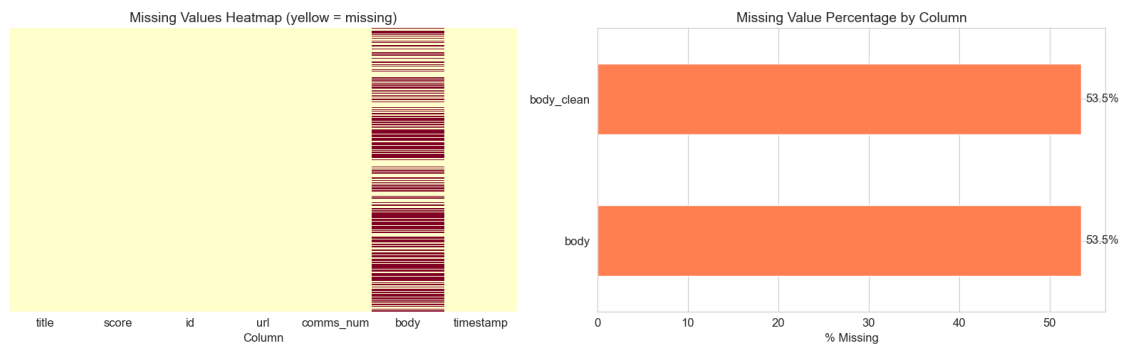
1.3.6 Preprocessing decision table (issue -> method -> justification)

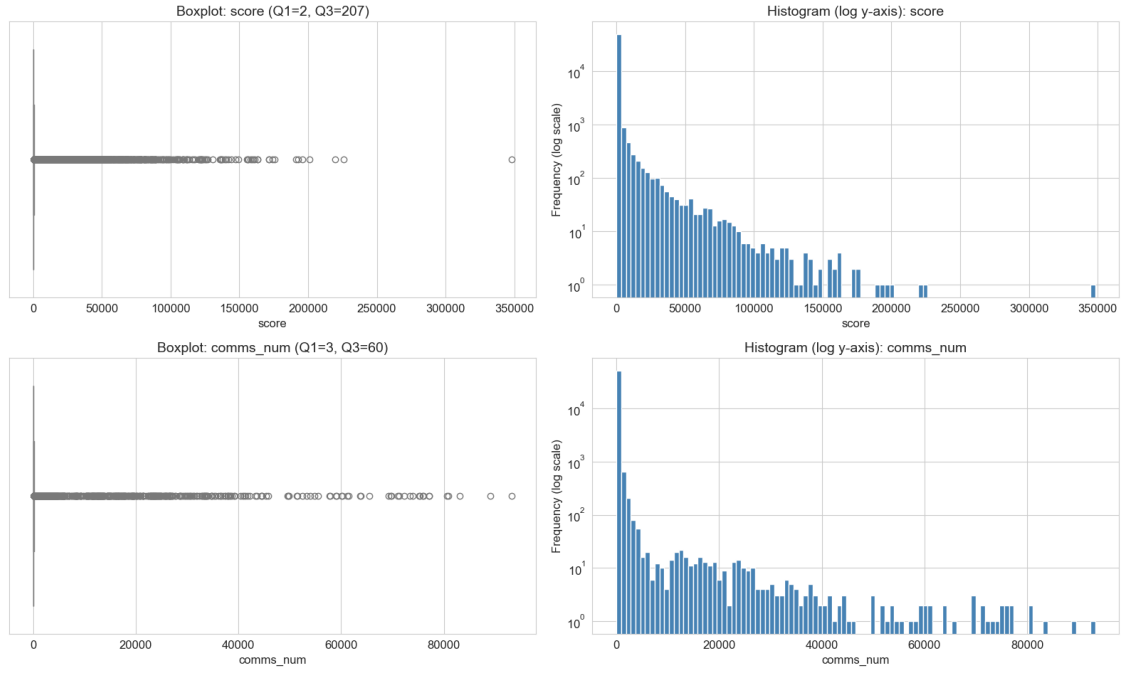
Issue	Method	Justification	Evidence
Incorrect/variable types	Strict coercion + normalization	Prevent invalid downstream feature extraction	02_cleaning_policy_metrics.json
Duplicates	ID + full-row audit	Avoid silent inflation bias	02_cleaning_policy_metrics.json
Missing text	Structural treatment + indicators	Preserve representativeness while modeling missingness explicitly	02_cleaning_fig_01_missingness_c
Outliers	log1p, robust diagnostics	Reduce extreme leverage from viral tails	02_cleaning_fig_03_outlier_visu
Scale mismatch	z-score + min-max features	Support comparable model inputs	02_cleaning_fig_07_scaling_compa

```
[ ]: # Required cleaning metrics (quick reproducibility excerpt)
missing_body_pct = raw['body'].isna().mean() * 100
dup_id = raw['id'].duplicated().sum()
dup_row = raw.duplicated().sum()

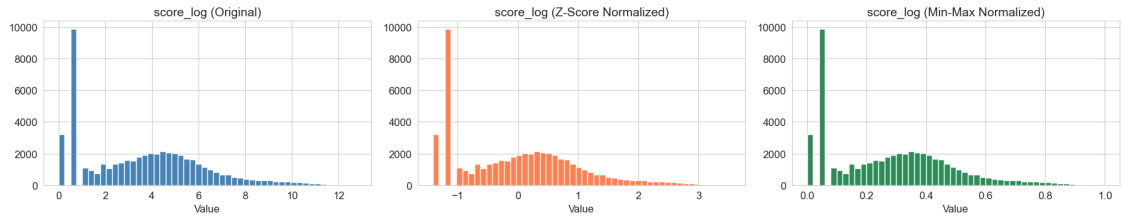
print('missing_body_pct:', round(missing_body_pct, 4))
print('duplicate_id_count:', int(dup_id))
print('duplicate_full_row_count:', int(dup_row))
```

1.3.7 Core Cleaning Figures





Effect of Different Scaling Methods



1.4 4. Exploratory Data Analysis (EDA)

The EDA stage combines descriptive views and inferential statistics to uncover nuanced behavior.

1.4.1 4.1 Descriptive + relational patterns

- Distributional skew and transformed distribution stabilization.
- Engagement coupling between score and comments.
- Temporal and post-type variation patterns.

1.4.2 4.2 Advanced statistical diagnostics

- Spearman correlation ($\rho = 0.7852$, $p < 0.001$) for monotonic engagement dependence.
- Kruskal-Wallis post-type test ($H = 5855.658$, $p < 0.001$) for group differences.
- Outlier-rate diagnostics ($|z| \geq 3$ rate $\sim 0.173\%$).

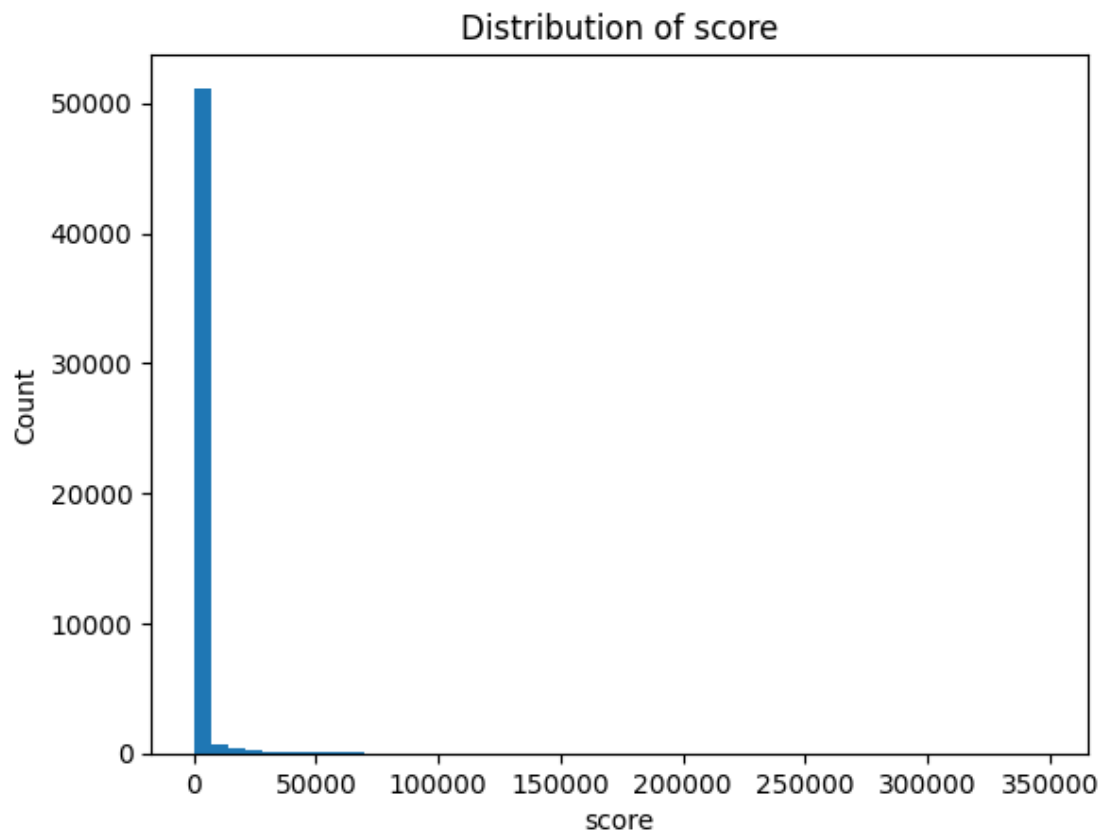
1.4.3 4.3 Interpretation

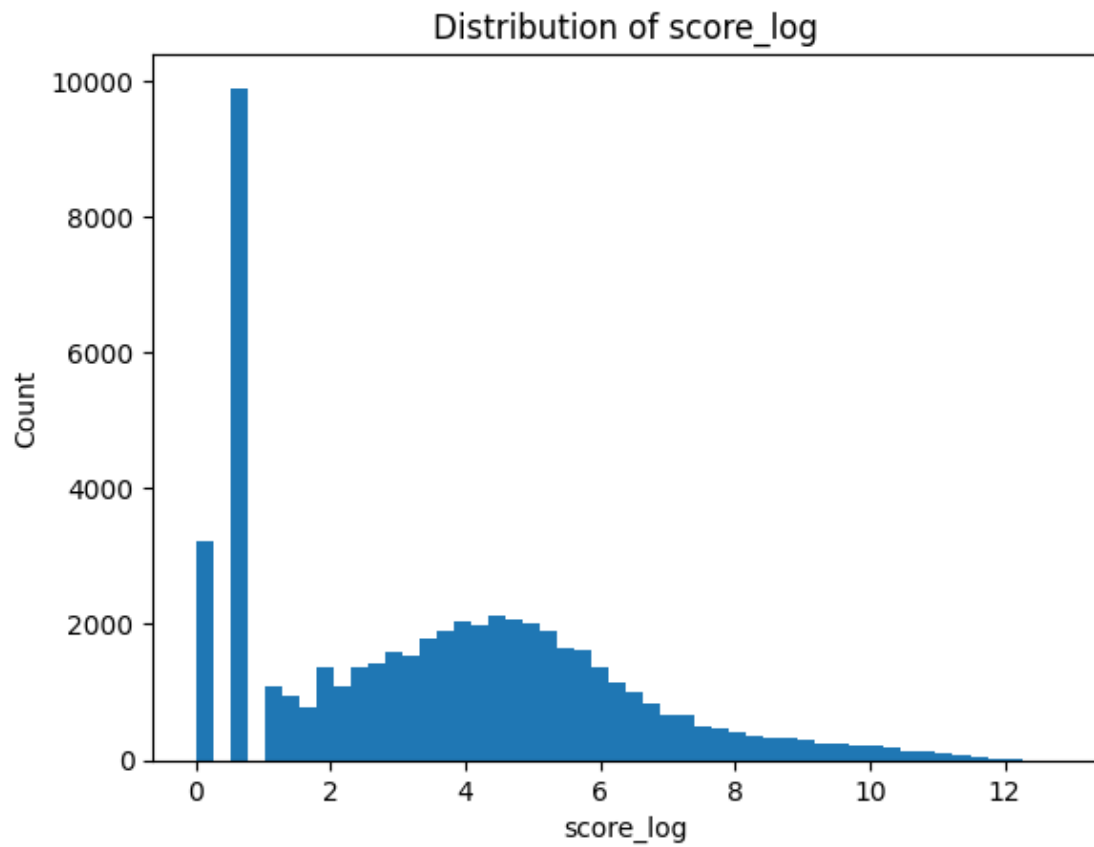
The data shows event-driven attention clustering: most posts are low-engagement, while rare viral spikes dominate influence, requiring robust transforms and caution in practical significance interpretation.

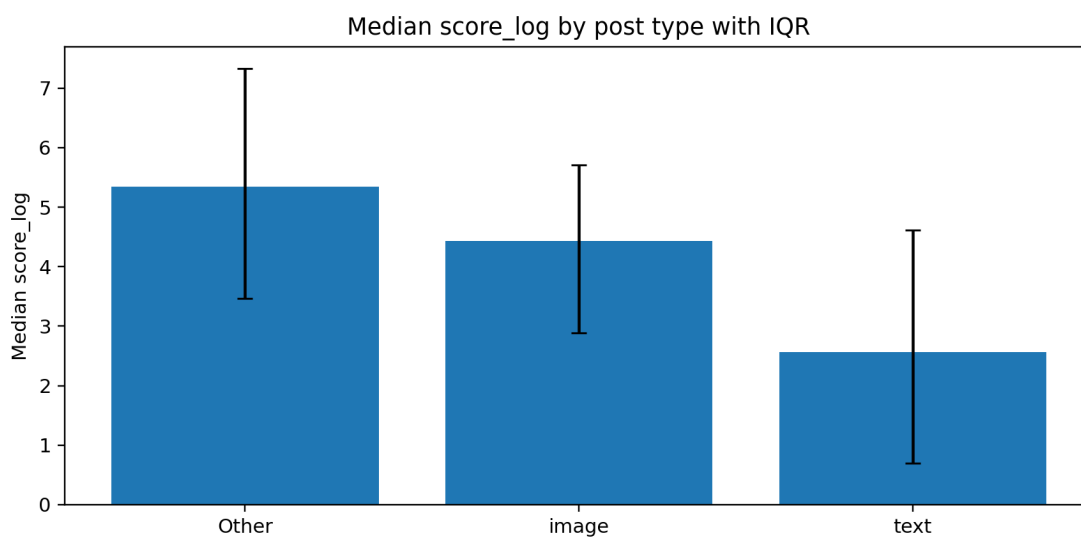
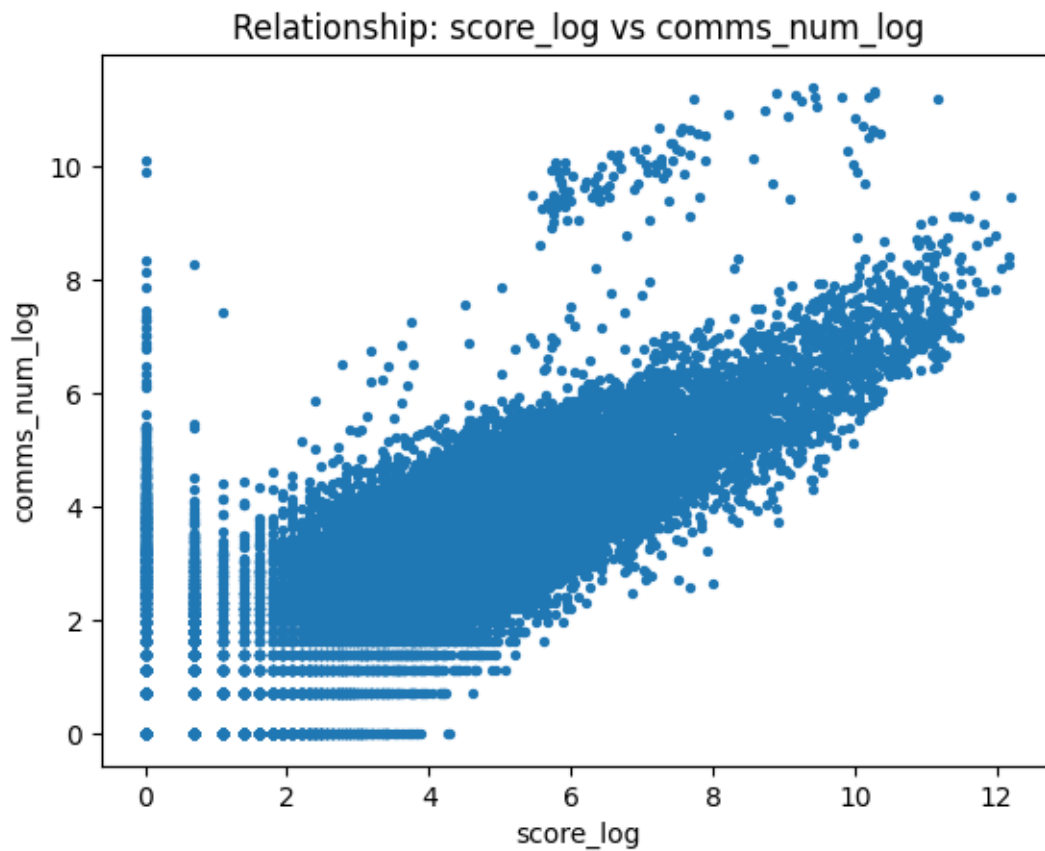
```
[ ]: import json
      from pathlib import Path

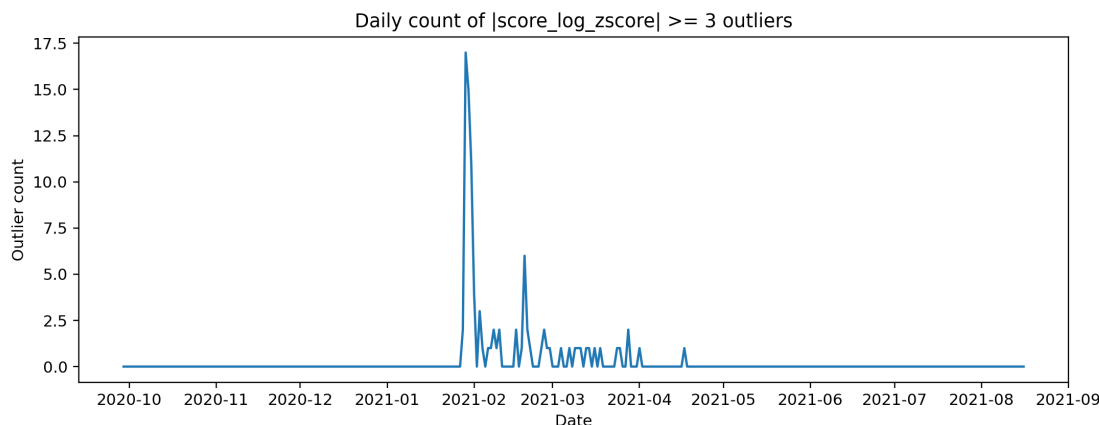
      stats_path = Path('../..//Project Workspace/Supporting Materials/Report Sources/
        ↳artifacts/json/03_eda_advanced_stats.json')
      eda_stats = json.loads(stats_path.read_text())
      print(eda_stats)
```

1.4.4 Core EDA Figures









1.5 5. Feature Engineering Process and Justification

Feature engineering extends baseline engagement predictors with text-derived signals to evaluate incremental predictive utility.

1.5.1 5.1 Engineered features

- Sentiment score from title lexicon rules.
- Topic-probability features via LDA topic modeling.
- Encoded/normalized predictors from preprocessing stage.

1.5.2 5.2 Ablation (before/after utility)

Model	Feature Count	ROC-AUC	Avg Precision	F1@0.5
Baseline	6	0.999984	0.999710	0.876950
Baseline + Sentiment	7	0.999984	0.999710	0.876950
Baseline + Sentiment + Topics	12	0.999975	0.999546	0.884615

Interpretation: - Sentiment alone produced negligible lift under this target definition. - Topic features improved threshold performance (+0.007666 absolute F1 lift), showing context adds practical decision utility.

1.5.3 5.3 Limitations

- Virality label is score-derived and may induce leakage-like ease.
- Lexicon sentiment has sarcasm/context limitations.
- Topic features improve performance but reduce direct single-post interpretability.

Source JSON: Project Workspace/Supporting Materials/Report Sources/artifacts/json/05_feature_ablation_table.json

```
[ ]: ablation_path = Path('../..//Project Workspace/Supporting Materials/Report_
↳Sources/artifacts/json/05_feature_ablation_table.json')
ablation = json.loads(ablation_path.read_text())
```



```
for row in ablation['models']:
    print(row)
```

1.5.4 Workflow Architecture (One Code File Visibility)

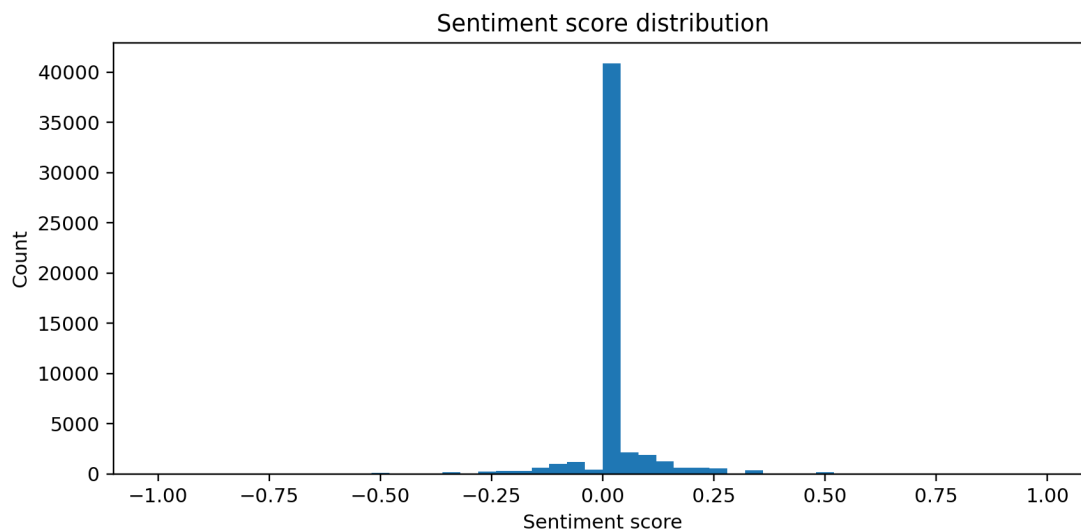
The full workflow is implemented in one canonical script: - `../Code Files/Project1_Full_Workflow_Code.ipynb` (canonical review code file) - `../Code Files/full_workflow.py` (backup script)

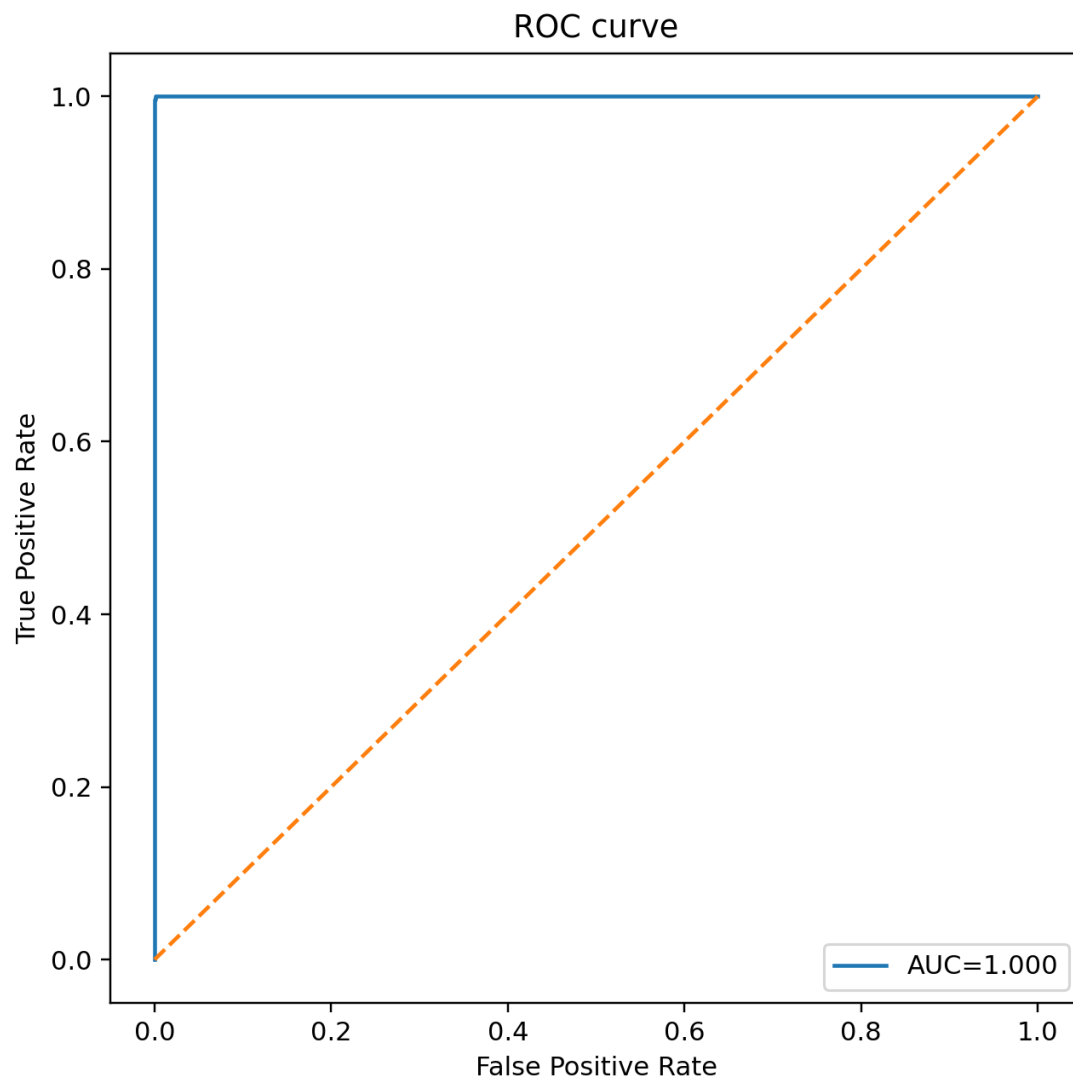
Pipeline stages: 1. Load and validate raw schema. 2. Cleaning/preprocessing transforms and feature preparation. 3. EDA figure/stat generation. 4. Feature diagnostics and ablation evaluation. 5. Export artifacts and summary JSON.

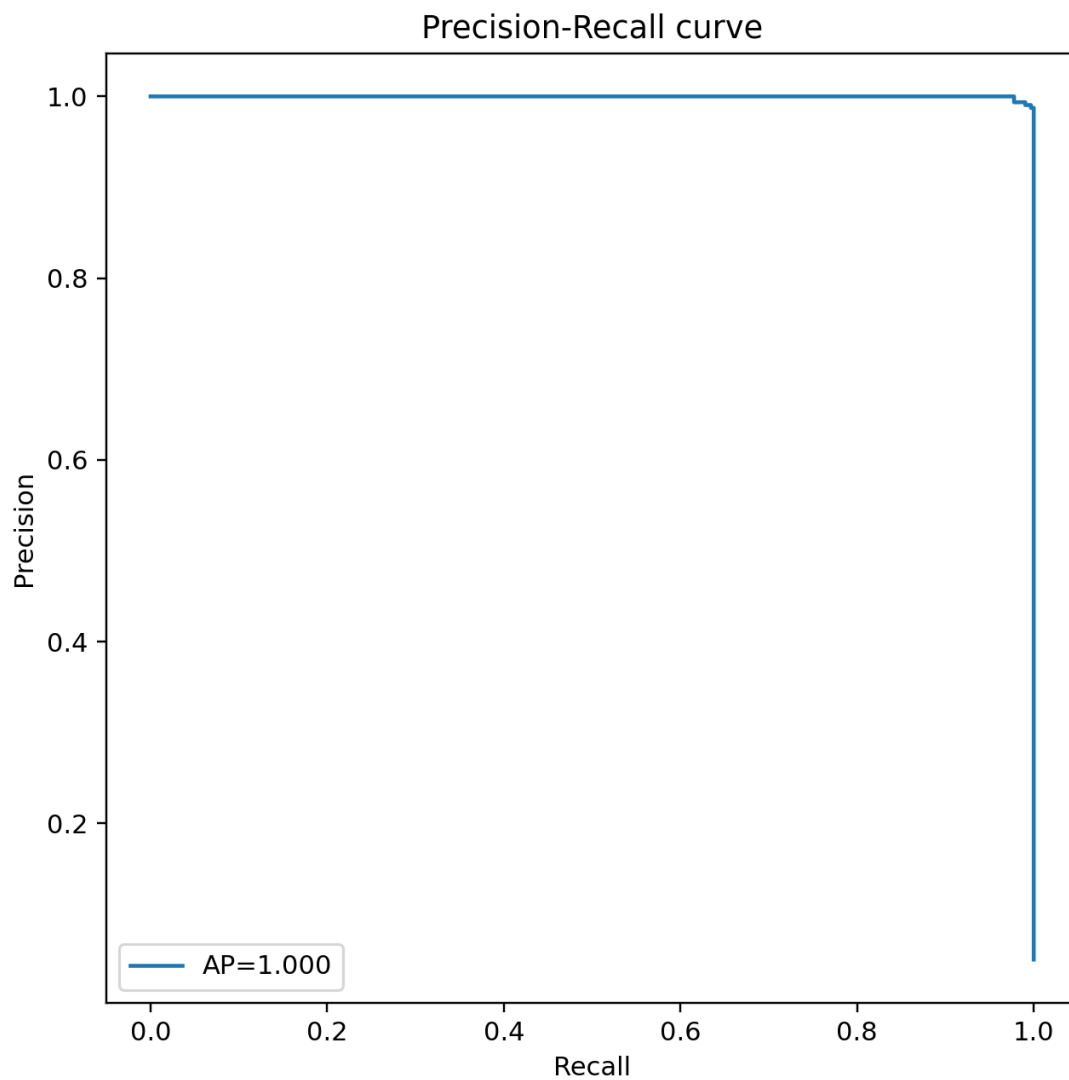
CLI example:

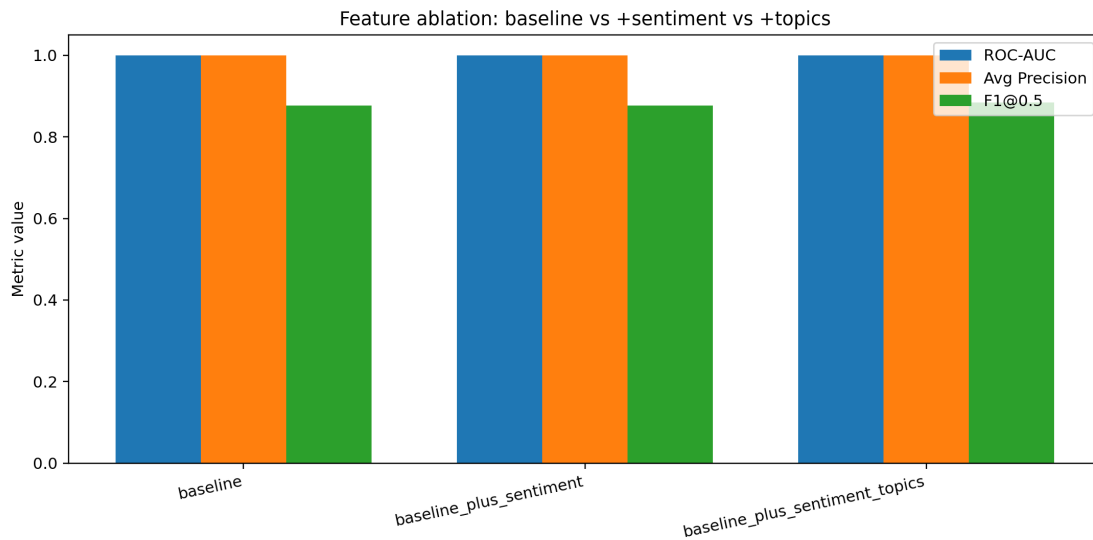
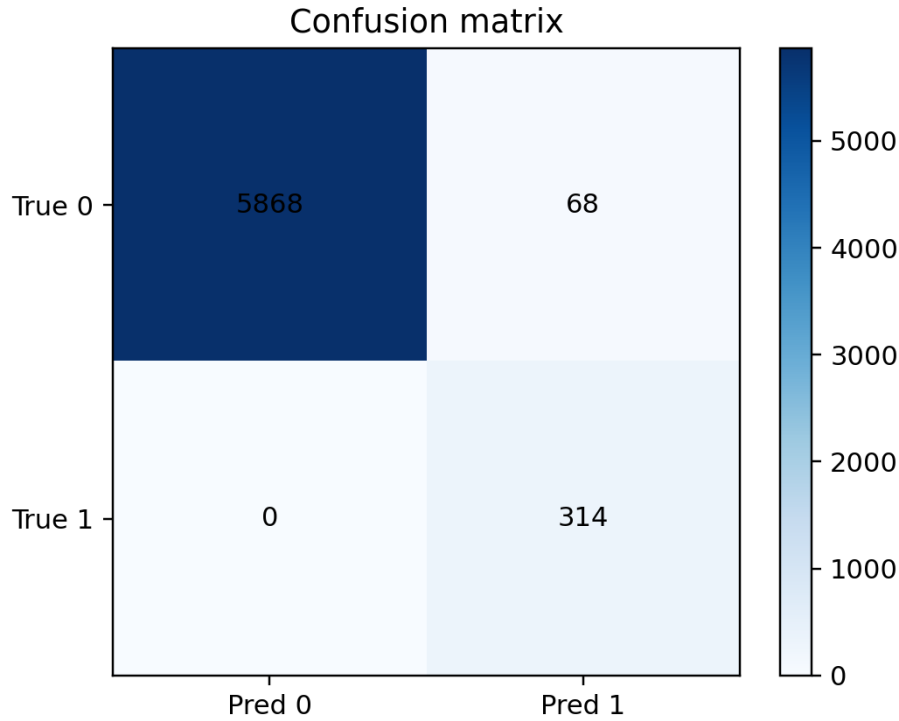
```
jupyter nbconvert --to notebook --execute "Project Deliverables/Code Files/Project1_Full_Workf
python3 "Project Deliverables/Code Files/full_workflow.py" --raw "Project Deliverables/Datasets"
```

1.5.5 Core Feature Figures









1.6 6. Summary of Key Findings

- Data complexity is high due to mixed structures, structural missingness, and heavy-tailed engagement outcomes.
- Cleaning addressed all major inconsistency types with explicit, auditable policy choices.
- EDA confirmed strong engagement dependence and statistically significant post-type effects.
- Feature engineering showed measurable threshold-lift from topic features.
- The one-file workflow is reproducible and supported by figure/JSON evidence artifacts.

Conclusion: the project satisfies full pipeline expectations with advanced-level evidence depth and transparent reproducibility.

1.6.1 Figure-to-Finding Traceability (Core Claims)

Claim ID	Core finding	Figure evidence	JSON evidence
C1	Structural missingness is high (~53.49% body missing) and must be handled explicitly.	02_cleaning_fig_01_missingness_diagnostics.png	Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig_01_missingness_diagnostics.png
C2	Engagement metrics are monotonic with strong dependence between score and discussion volume.	03_eda_fig_10_score_comments_scatter.png	Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_fig_10_score_comments_scatter.png
C3	Score distribution has heavy tails with a small extreme outlier fraction.	03_eda_fig_01_score_distribution.png, 03_eda_fig_17_daily_outlier_fraction.png	Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_fig_01_score_distribution.png, 03_eda_fig_17_daily_outlier_fraction.png
C4	Feature enrichment with topics improves threshold utility beyond baseline/sentiment-only.	05_feature_ablation_comparison.png	Workspace/Supporting Materials/Report Sources/artifacts/json/05_feature_ablation_comparison.png
C5	Classifier discrimination is high and supported by ROC/PR diagnostics.	04_feature_fig_04_roc_curve.png, 04_feature_fig_05_pr_curve.png	Workspace/Supporting Materials/Report Sources/artifacts/json/04_feature_fig_04_roc_curve.png, 04_feature_fig_05_pr_curve.png

1.7 7. Challenges Faced and Future Recommendations

1.7.1 Challenges

- Structural missingness in body text and noisy social-media language.
- Heavy tails and rare viral events complicating stable modeling.
- Label-definition caveat due to score-derived virality target.

1.7.2 Recommendations

- Move toward forward-looking targets and temporal validation splits.
- Add richer contextual NLP and sarcasm-aware sentiment approaches.
- Integrate external market variables for stronger cross-domain interpretation.

1.8 8. GitHub Repository Link

- <https://github.com/ZemingLiang/STAT-5243-Project-1-Team-2>

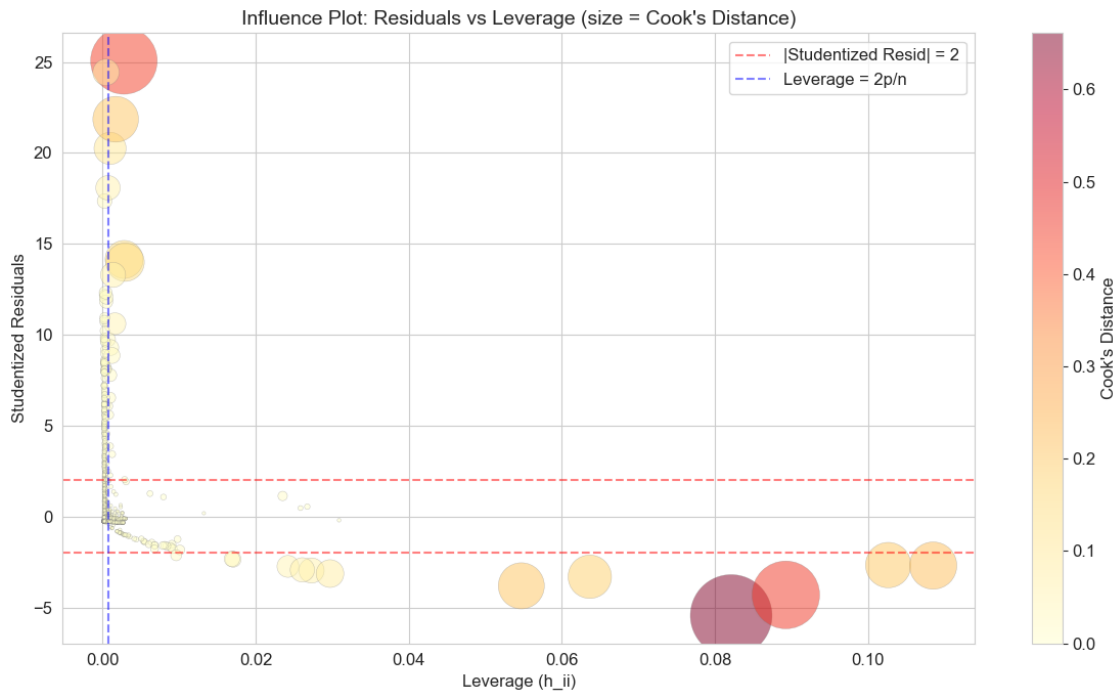
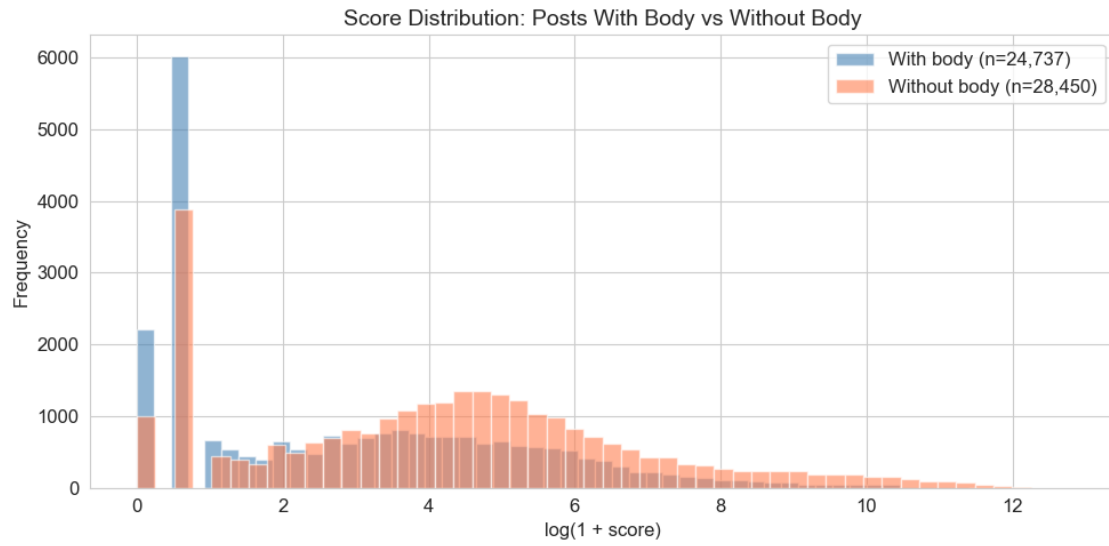
1.9 9. Each Member’s Contribution

- **Zeming Liang:** Data cleaning and preprocessing; creation/normalization of reddit_wsb.csv, reddit_wsb_cleaned.csv, and Cleaning-and-Preprocessing.ipynb; final repo organization and PNG/JSON/output integration.
- **Zuer Weng:** EDA analysis, advanced statistical diagnostics, and EDA branch artifacts.
- **Duoli Chen:** Report-writing narrative sections (introduction, acquisition, summary, challenges) and narrative refinement.
- **Isaac Beers:** Feature engineering workflow, viral-classification diagnostics, and model interpretation components.

1.10 Appendix A. Full Evidence Figure Index (Inside Same PDF)

This appendix contains additional supporting visuals so the report remains one self-contained grading artifact.

Figure ID	Section	Purpose
02_cleaning_fig_02_score_by_gender.png	Cleaning	Supporting diagnostic evidence
02_cleaning_fig_04_influence_plot.png	Cleaning	Supporting diagnostic evidence
02_cleaning_fig_05_qq_normality.png	Cleaning	Supporting diagnostic evidence
02_cleaning_fig_06_spearman_correlation.png	Cleaning	Supporting diagnostic evidence
03_eda_fig_02_comms_distribution.png	EDA	Supporting diagnostic evidence
03_eda_fig_04_comms_log_distribution.png	EDA	Supporting diagnostic evidence
03_eda_fig_05_title_length_distribution.png	EDA	Supporting diagnostic evidence
03_eda_fig_06_hour_distribution.png	EDA	Supporting diagnostic evidence
03_eda_fig_07_post_type_lump_bar.png	EDA	Supporting diagnostic evidence
03_eda_fig_08_post_type_bar.png	EDA	Supporting diagnostic evidence
03_eda_fig_09_day_of_week_bar.png	EDA	Supporting diagnostic evidence
03_eda_fig_11_title_score_scatter.png	EDA	Supporting diagnostic evidence
03_eda_fig_12_score_hour_trend.png	EDA	Supporting diagnostic evidence
03_eda_fig_13_comments_hour_trend.png	EDA	Supporting diagnostic evidence
03_eda_fig_14_score_dow_trend.png	EDA	Supporting diagnostic evidence
03_eda_fig_15_daily_score_trend.png	EDA	Supporting diagnostic evidence
03_eda_fig_16_corr_heatmap.png	EDA	Supporting diagnostic evidence
03_eda_fig_18_hour_day_heatmap.png	EDA	Supporting diagnostic evidence
04_feature_fig_02_topic_entropy_distribution.png	Feature Engineering	Supporting diagnostic evidence
04_feature_fig_03_dominant_topics_distribution.png	Feature Engineering	Supporting diagnostic evidence
04_feature_fig_07_pred_prob_distribution.png	Feature Engineering	Supporting diagnostic evidence
04_feature_fig_08_top_coefficients.png	Feature Engineering	Supporting diagnostic evidence



Normality Assessment: Before vs After Log Transform

